

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Heavy metals: lead, mercury - Set 1

Mark scheme

AQA spec 3.4.3.2.11; Pollution; textbook pp. 275-279

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $(15.0 - 2.2) / 15.0 \times 100$ 1 = 85.3% (accept 85%) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. Do not credit • A bare wrong answer with no working.	2

Question	Marking guidance	Marks
01.2	1 mark each 1 mark for each of the 3 points below. 1 Added as an anti-knock agent to smooth combustion and reduce engine wear 1 Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food 1 Lead is a neurotoxin that bioaccumulates – chronic exposure damages the nervous system / brain development in children Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. particulates → fine solid particles / smoke and dust	3
01.3	Any 2 of 5 1 mark each for any 2 of the 5 points below, in any order, maximum 2. 1 Replacement of lead water pipes / lead solder with copper / tin-based solder – phosphoric acid dosing of water supplies forming insoluble lead phosphate 1 Phasing out of lead-based paint / safer paint removal 1 Lead-free electrical solder 1 Restrictions on lead fishing weights / shotgun pellets 1 Industrial emission controls Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. - quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
01.4	1 mark each 1 mark for each of the 2 points below. 1 Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe 1 This prevents lead dissolving into the water Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	1 mark for the point below. 1 Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.		

Question	Marking guidance	Marks
02.1	1 mark each 1 mark for each of the 4 points below. 1 Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury 1 Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes 1 It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water 1 Local people ate large amounts of fish (critical group) so received toxic doses Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. critical group → the people most at risk / most heavily exposed Do not credit 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
02.2	1 mark each 1 mark for each of the 3 points below. 1 Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) 1 It crosses the placenta and harms the developing unborn baby 1 Mercury is a teratogen, preventing normal gene expression / development Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
02.3	1 mark each 1 mark for each of the 3 points below. 1 Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs 1 Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut 1 Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs – liposoluble so most toxic and bioaccumulates Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.		

Question	Marking guidance	Marks
03.1	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Disposal of items containing mercury: batteries, thermometers, fluorescent lights 1 Chemical plants producing chlorine using mercury electrodes 1 Combustion of coal 1 Gold mining (amalgamation) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
03.2	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) 1 Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters 1 Disposal / storage of wastes at high pH to reduce solubility 1 Encapsulation of mercury-containing hazardous waste in cement Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
03.3	1 mark each 1 mark for each of the 2 points below. 1 Most heavy metals are more soluble at low pH / less soluble at high pH 1 So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. leached → nutrients or ions washed down out of the soil / washed away by water Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Question	Marking guidance	Marks
03.4	1 mark each 1 mark for each of the 2 points below. 1 The effects of two pollutants interact to produce a different (greater) effect than either alone 1 Cadmium and zinc – ozone and sulfur dioxide Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - Any appropriate named example, including ones that are not listed here. synergism → two pollutants together doing more damage than the sum of each alone Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	
04.1	Calculation 1 mark for the correct answer; working does not have to be shown. 1 $180 / 1.2 = 150$ Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (times) is already printed on the answer line. Do not credit - A bare wrong answer with no working. 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
04.2	1 mark for the point below. 1 Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1

Question	Marking guidance	Marks
04.3	1 mark each 1 mark for each of the 4 points below. 1 Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium (1.2 vs 0.05 mg kg^{-1}) 1 Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years 1 Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) 1 Rice is the staple food so residents receive a large dose – village B has no exposure and no disease (correlation supports causation) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. correlation → a relationship in which the two variables change together causation → one variable actually causing the change in the other Do not credit - a correlation presented as proof of cause - say that it suggests a link that further work would have to test 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
04.4	Any 2 of 4 1 mark each for any 2 of the 4 points below, in any order, maximum 2. 1 Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble 1 Cover / cap spoil heaps to prevent leaching and dust 1 Recycle nickel-cadmium batteries / hazardous waste landfill 1 Phytoremediation of contaminated paddy soils Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. leaching → nutrients or ions washed down out of the soil / washed away by water Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
04.5	1 mark each 1 mark for each of the 2 points below. 1 Manufacture may release cadmium in effluents / dust 1 Panels have a limited life – disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. incineration → burning waste, usually with energy recovery Do not credit 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
05.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 B	1
05.2	1 mark each 1 mark for each of the 3 points below. 1 TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations 1 It bioaccumulated in shellfish and could pass to humans eating them 1 It was assumed dilution would make it harmless (illustrates the need for the precautionary principle) – now prohibited and replaced by copper Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	3
05.3	1 mark each 1 mark for each of the 3 points below. 1 In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river 1 In the river it is oxidised to insoluble iron oxide (orange sediment) – the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms 1 Control: drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river – the solid iron is removed periodically Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • leached → nutrients or ions washed down out of the soil / washed away by water	3
05.4	Any 2 of 3 1 mark each for any 2 of the 3 points below, in any order, maximum 2. 1 Lead-acid car batteries (recycled) 1 Lead flashing on roofs / windows 1 Radiation shielding Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Question	Marking guidance	Marks
05.5	1 mark for the point below. 1 They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
06.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 C: Organic (methyl) mercury Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
06.2	1 mark each 1 mark for each of the 2 points below. 1 Anaerobic microbes converted the inorganic mercury into organic methyl mercury 1 Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
06.3	1 mark each 1 mark for each of the 2 points below. 1 Most heavy metals are less soluble at high pH 1 So they remain immobile and are not leached into water or absorbed by organisms Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. leached → nutrients or ions washed down out of the soil / washed away by water Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	